

References and Resources

“Make Your Own MCU Boards”, Nathan Jones

Circuits

- Nucleo-F042K6
 - [Dev board product page](#)
 - [MCU product page](#)
 - [Datasheet](#)
 - [Hardware development guide](#)
- PIC16F18446 Curiosity Nano
 - [Dev board product page](#)
 - [MCU product page](#)
 - [Datasheet](#)
- [STM32F103C8T6 Breakout Board](#) (nathancharlesjones)
- Useful circuits: [Electronics Cookbook](#) (Simon Monk)
- Clocks
 - [Oscillator design appnotes](#)
 - [Myths around load capacitance – how to choose the right capacitors](#) (IQD)
 - [How a few picofarads can ruin your timing accuracy](#) (IQD)
 - [What effect does selecting the wrong load capacitance for a crystal have?](#) (StackExchange)

Parts

- [AdaWiki PartSelector](#)
- [JLC PCB parts library](#)
- Breakout boards
 - [SOIC/TSSOP](#) and [QFN/QFP](#) (Adafruit)
 - [SMT Breakout Boards](#) (marsairforce)
 - [Surfboards 6000 series](#) (Capital Advanced)
 - [SMT to DIP adapters](#) (Schmartboard)
- Power
 - [Adafruit USB breakout boards](#)
 - [Breadboard-friendly DC jack](#) (Adafruit #373)
- Clock
 - [8MHz through-hole oscillator](#) (Digi-Key)
- Programming
 - [Debug edge](#)
 - [Debug Edge adapter: SWD to J-Link Edu Mini with USB power](#) (nathancharlesjones)

Layout

- Breadboards/Protoboards
 - [Breadboard layout sheets](#) (Instructables)
- PCBs
 - [JLC PCB capabilities](#)
 - JLC PCB design rules files
 - [Eagle](#)
 - [KiCAD](#)
 - [High speed PCB design ebook](#) (Sierra Circuits)
 - [11 Best High-Speed PCB Routing Practices](#) (Sierra Circuits)
 - [4PCB Trace Width Calculator](#)
 - [SMPS Voltage Clearance Calculator](#)

Construction

- Prototyping boards
 - Breadboards: [Half-size](#) | [Full-size](#) (Adafruit)
 - [Breadboard PCB](#) (nathancharlesjones)
 - [Stripboard PCB](#) (nathancharlesjones)
- SMT soldering
 - [EZ Make Oven](#) (Adafruit)
 - [Introduction to the Manually-Controlled Toaster Oven Reflow](#) (All About Circuits)
 - [Surface Mount Soldering with a Toaster Oven](#) (Freertronics)
- [JLC PCB assembly](#)
 - [Example BOM file](#)
 - [Example pick and place file](#)
- [MakerFabs PCB assembly](#)